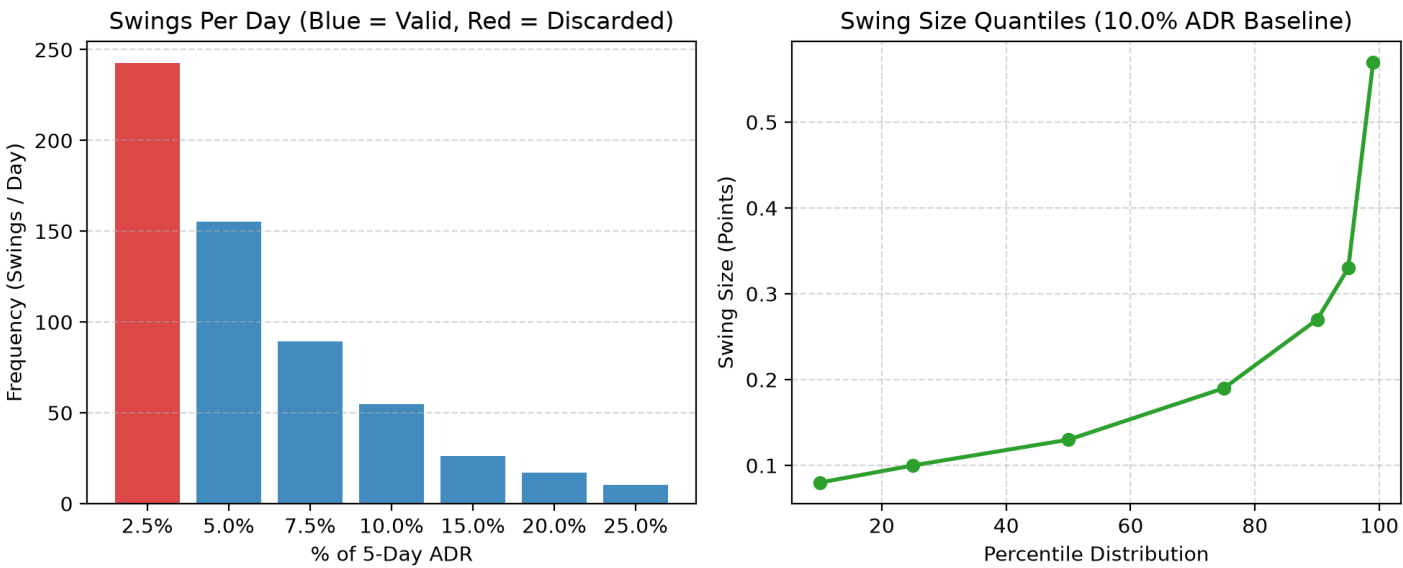


Statistical Swing Analysis Report: ITUB4 (1min)

5-Day Average Daily Range (ADR_5d): 0.74 pts

ITUB4 (1min) - Swing Distribution & Sensitivity Analysis



Parameter Sweep & Noise Filtering Matrix

Pct ADR	Threshold Pts	Total Swings	Swings Per Day	Avg Bars Per Swing	Mean Size	Median Size	P75 Size	P90 Size	P95 Size	Status
2.5%	0.02	39298	242.6	2.7	0.07	0.05	0.08	0.12	0.16	DISCARDED
5.0%	0.04	25126	155.1	3.7	0.09	0.07	0.11	0.15	0.2	VALID
7.5%	0.06	14444	89.2	5.7	0.12	0.1	0.14	0.21	0.26	VALID
10.0%	0.07	8873	54.8	8.7	0.16	0.13	0.19	0.27	0.33	VALID
15.0%	0.11	4235	26.1	17.1	0.24	0.2	0.28	0.4	0.5	VALID
20.0%	0.15	2734	16.9	25.9	0.3	0.25	0.35	0.51	0.64	VALID
25.0%	0.19	1666	10.3	41.8	0.39	0.32	0.45	0.68	0.86	VALID